Hard braking Math

Assuming 1,500 psi is the pressure at which the driver breaks,

Brake Pressure Sensor Range
0–2000 psi
Output Voltage Range
.5 – 4.5

$V_{out\ pressure\ sensor} = 4.5 \cdot \frac{\text{brake pressure psi}}{2000}$

$4.5 \cdot \frac{1500}{2000} = 3.375\ V$ = hard braking threshold

$V = IR$
$P = IV$
$P = \frac{V^2}{R}$

$V_{out} = V_{in}\frac{R_2}{R_2+R_1}$

$25{,}000\ \Omega = R_1 + R_2$

skipping a few steps

$3.375 = \frac{R_2}{2.5\times10^3}$

$3.375\times10^5 \div 2.5 = R_2 = 8437.5\ \Omega$

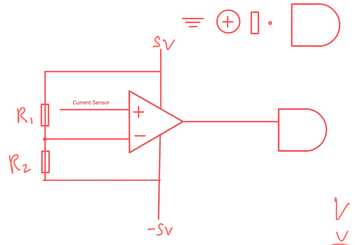
$V_{out} = \frac{R_2}{R_1+R_2}$

$R_1 = 16{,}562.5\ \Omega$

Hard braking Resistor Voltages

HASS 300-S stuff

High Current mat



$U_{Ref} = 2.5V$ 5000 watts => 5000 joules/s

Nominal battery voltage = 300 V nominal

$I_{threshold} = \frac{5000\ J}{300\ V} = 16.6\bar{7}$ Amperes

$V_{out\ current\ sensor} = U_{ref} + .625V \times (I_P / I_{PN})$

$V_{out} = 2.5V + .625V \times (16.67/300) = 2.534V$

Looks barely different from V_{out} when the current sensor reads 0 current out :)

$V = IR$
$\frac{V}{R} = I$

$V_{out} = V_{in}\frac{R_2}{R_2+R_1}$

2.534

$R_2 + R_1 = 2.5\times10^3\ \Omega$

$R_1 = 2.5\times10^3\ \Omega - R_2$

$V_{out} = \frac{R_2}{R_2+(2.5\times10^3\ \Omega - R_2)} = \frac{R_2}{2.5\times10^3} = 2.534$

$V_{out} = 2.534\ V$

$R_2 = 2.534 \cdot 2.5\times10^3 = 6335\ \Omega$

$R_1 = 3835$

High Current Resistor Voltages